Nkanyiso Owethu Ndlovu

0223062870 | nkanyisoowethu@gmail.com | linkedin.com/in/nkanyiso-ndlovu-46a30821b | github.com/NkaLight

Education

University of Otago Dunedin, NZ
Bachelors of Applied Science, Double major in Computer Science & Software Engineering *Feb 2023 – Dec 2025*

Projects

Study assistance App (NotePilot) | *Flask API,Next.js, Typescript* | **GitHub Link** July 2025 – Present

- I gained exposure to project management tools such as Taiga, which helped our team of 5 stay on schedule over the course of the project.
- Deployed a Python RAG microservice on AWS ECS, connected it to the Next.js main application(Vercel) through a private internal API secured with a shared key, ensuring that only the Next.js could trigger the RAG pipeline. This reduced hallucinations by retrieving semantically relevant chunks from the database and also allowed the user easily spot hallucinations as the chatbot had to reference sections in the user's uploaded lecture material.
- To ensure stability of the codebase during a major refractor I implemented end to end testing in playwright.
- To allow users to reset their password I implemented password reset flow by integrating a free mail service Mailjet, with a one-time cryptographic key sent to the via email, allowing the user to reset their password securely.

Snakes - Genetic Algorithms Project | *Python, Numpy, ML* | **GitHub Link** July 2025 – Nov 2025

- I chose a single-layer perceptron over a deeper network because the cap on 500 generations was insufficient to exploit the extra capacity. The smaller search space made crossover and mutation more effective and the training dynamics easier to debug
- I chose tournament selection as my selection method over roulette wheel to maintain genetic diversity and avoid premature convergence, while using elitism to preserve the fittest agent each generation. This balanced exploration and exploitation of the solutions.
- After each parameter change (mutation rate, mutation strength, fitness weights), I collected average fitness score across generations to diagnose premature convergence or noise, using that data to incrementally refine the algorithm.
- Achieved 100% win rate against the random baseline, however the agents placed lower on the leaderboard against other agents due to an overly extensive fitness function, teaching me the trade-off between reward precision and generalization.

ANDIE - A Non-Destructive Image Editor | *Java, Swing, JUnit, Git* | **Github Link** Jun 2024- Nov 2024

- When building a desktop editor for a university team project, I took ownership of brightness/contrast filter, random scattering filter, and the export module, allowing users to precise light-control and a reliable one-click PNG output
- To make the tool usable by an international audience I implemented internationalization. This at the moment is scoped to English and Te Reo Māori.
- As part of a five-developer team using Git, I engaged in pair programming, code reviews with the team and merged feature branches which kept integration smooth and the project was delivered on schedule.

Experience

Research Assistant Studentship Nov 2025 – Jan 2026
University of Otago *Dunedin, NZ*

- Assisted with empirical research by performing software repository mining(Python script), interacting with the GitHub API, managing the 5000 requests per hour rate limit constraint and I stored the collected data in relational database for reproducible analysis.
- Designed and implemented the schema where the data was stored relationally and used it to anaylze how developers are augmenting their CI with LLMs. Through the use of SQL queries and data visualization tools such as R and python matplotlib, I found that corrective maintenance workflow files increased when developers started using LLMs to augment their CI.

Waiter Aug 2023 – Nov 2025
Vedura *Dunedin, NZ*

- I provided exceptional customer service and when we were too busy to serve everyone immediately I learned to manage customer expectations early, resulting in smoother service.
- I trained incoming staff on the overall operations the restaurant and front of house resulting in a better more faster on boarding period for new staff.

Technical Skills

Languages: Java, Python, SQL (Postgres), JavaScript, HTML/CSS, R
Frameworks: React, Next.js, Node.js, Flask, JUnit, Tailwind CSS, FastAPI
Developer Tools: Git, Docker, VS Code, Visual Studio, AWS
Libraries: pandas, NumPy, Matplotlib

References

Sherlock Lickorish Associate Professor, School of Computing, University of Otago
sherlock.licorish@otago.ac.nz

Daniel Alencar da Costa Senior Lecturer, School of Computing, University of Otago
danielcalencar@otago.ac.nz